Subjective Handwritten Answer Evaluation using NLP

Submitted in partial fulfilment of the requirements
of the degree of

Bachelor of Engineering

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(2025-2026)

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CERTIFICATE

This is to certify that the project entitled **"Subjective Handwritten Answer Evaluation using NLP"** is a bonafide work of **Sami Ansari (23106093), Jeet Jain (23106029), Swayam Bhoir (23106025), Lucky Gupta (23106118)** submitted to the University of Mumbai in partial fulfilment of the requirement for the award of the degree of **Bachelor of Engineering** in **Computer Science & Engineering (Artificial Intelligence & Machine Learning**)

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ABSTRACT

Evaluating handwritten subjective answers is a time-consuming and inconsistent task due to human bias, fatigue, and subjective interpretation. To overcome these challenges, the project "Subjective Handwritten Answer Evaluation using NLP" supports Sustainable Development Goal 4 – Quality Education, which promotes fair, inclusive, and effective learning.

The proposed system, Answer Check, is an AI-based web application that automates the evaluation of descriptive handwritten answers using Natural Language Processing (NLP) and Optical Character Recognition (OCR). The Gemini Vision API extracts text from scanned or photographed answer sheets, while pre-processing techniques such as tokenization, stop-word removal, and lemmatization (via spaCy and NLTK) refine the data. The system uses TF-IDF vectorization and cosine similarity to compare student responses with model answers, focusing on semantic understanding rather than keyword matching.

Developed using Python (FastAPI) and React (Vite), Answer Check ensures accuracy, fairness, and consistency in evaluation. Additional modules like keyword detection and tone analysis further enhance grading reliability. Experimental results show improved grading consistency and reduced educator workload, enabling teachers to focus on teaching and mentoring.

Thus, Answer Check promotes efficient, unbiased assessment and contributes to the advancement of Quality Education through AI-driven automation. Can be used for large-scale online education platforms (MOOCs, digital universities, etc.) to evaluate descriptive answers efficiently

Keywords: Quality Education, SDG 4, NLP, OCR, FastAPI, Gemini Vision API, TF-IDF, Cosine Similarity, Automated Grading.

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ABBREVIATION

AI .....*Artificial Intelligence*

BERT.....*Bidirectional Encoder Representations from Transformers*

CNN.....*Convolutional Neural Networks*

HTR .....*Handwritten Text Recognition*

IDE .....*Integrated Development Environment*

ML .....*Machine Learning*

NLP.....*Natural Language Processing*

OCR.....*Optical Character Recognition*

OMR .....*Optical Mark Recognition*

TF-IDF.....*Term Frequency–Inverse Document Frequency*

Chapter 1

Introduction

Education has always been at the core of human progress, and evaluation remains one of its most essential components. Assessments not only measure a learner's grasp of concepts but also guide educators in identifying areas that require improvement. Among the various forms of assessment, subjective or descriptive answer evaluation stands out as one of the most challenging tasks. Unlike multiple-choice or objective questions, subjective answers demand interpretation, critical thinking, and contextual understanding, all of which require human-like judgment. Traditionally, this evaluation is carried out manually by teachers, who must read, analyse, and assign marks to each response based on relevance, coherence, and completeness. This process, though effective, becomes increasingly labour-intensive, inconsistent, and time-consuming as class sizes and exam volumes grow.

Manual evaluation is also subject to human limitations and biases. Factors such as fatigue, mood, time pressure, or individual interpretation can lead to inconsistent grading across evaluators or even by the same evaluator at different times. In large-scale examinations, ensuring uniformity and fairness in assessment becomes nearly impossible. This has led to a growing demand for automated systems that can provide objective, reliable, and scalable solutions for grading subjective answers, particularly those written by hand.

Recent technological advances have made it feasible to tackle this problem using Artificial Intelligence (AI), Natural Language Processing (NLP), and Optical Character Recognition (OCR).

OCR enables computers to accurately recognize and digitize text from scanned or photographed handwritten answer sheets, while NLP techniques allow machines to understand and evaluate the semantic content of that text. Together, these technologies open new avenues for automating the evaluation of handwritten subjective responses with accuracy and fairness.

The project titled “Subjective Handwritten Answer Evaluation using NLP” addresses this challenge through a web-based system named Answer Check, designed to automate the grading process while maintaining transparency and interpretability. The system uses the Gemini Vision API for text extraction, providing robust recognition across diverse handwriting styles and qualities. Once the handwritten content is converted into text, it undergoes text pre-processing, which includes tokenization, stop-word removal, and lemmatization using NLP libraries such as spaCy and NLTK. The processed text is then analysed semantically against a reference or model answer using TF-IDF vectorization and cosine similarity, implemented via Scikit-learn. This enables the system to compare meaning and context rather than relying solely on word overlap.

The backend of the system is built using Python and FastAPI, ensuring efficient handling of logic, requests, and API interactions, while the frontend is developed using React and Vite, offering a modern, responsive, and intuitive interface. Additional modules, including keyword detection and tone analysis, contribute to a comprehensive evaluation framework. The final scoring mechanism integrates multiple metrics: semantic similarity, keyword presence, and emotional tone; into a weighted model that approximates human evaluation standards.

The implementation of such a system brings numerous benefits. Educators can significantly reduce grading time while maintaining a high degree of consistency and fairness. Students, in turn, receive faster feedback and can better understand their performance through transparent evaluation criteria. Furthermore, the system’s scalability allows it to handle large datasets, making it suitable for academic institutions, competitive examinations, and e-learning platforms.

Ultimately, Answer Check exemplifies how cutting-edge AI and NLP technologies can be leveraged to revolutionize traditional educational processes. Beyond automating grading, it demonstrates the potential of technology to bridge the gap between human cognition and machine understanding. As the system evolves, it opens opportunities for future research and development in related domains. Thus, this project not only addresses a pressing challenge in educational assessment but also lays the groundwork for a new era of intelligent, fair, and efficient academic evaluation systems.

Chapter 2

Literature Survey

2.1 History

The quest to automate the evaluation of written responses dates back several decades, driven by the growing need for efficiency, accuracy, and consistency in educational assessment. Early systems were primarily designed to grade objective-type questions, such as multiple-choice or fill-in-the-blank items, using Optical Mark Recognition (OMR) technology. While these approaches significantly reduced manual workload, they were incapable of handling the complexity and subjectivity inherent in handwritten descriptive answers, which demand deeper comprehension of context, relevance, and reasoning. The journey toward automating subjective answer evaluation has evolved alongside developments in two major technological domains: Optical Character Recognition (OCR) and Natural Language Processing (NLP).

Early Stages (Pre-2000s):

The initial attempts at automating handwriting recognition were based on rule-based algorithms that analysed geometric features such as the shapes, curves, and strokes of handwritten characters. These systems, while pioneering, were highly sensitive to variations in individual writing styles, leading to frequent recognition errors. Similarly, early NLP applications for automated essay scoring relied heavily on keyword matching and pattern detection, offering only superficial text understanding. Such methods failed to assess the coherence, originality, and contextual accuracy of a student's response, limiting their practical value in real-world educational settings.

Machine Learning Era (2000s-2010s):

The rise of Machine Learning (ML) in the early 2000s marked a pivotal turning point. Researchers began employing statistical models and classifiers to identify patterns within handwriting data, resulting in more adaptive and reliable OCR systems. Machine-learning-based recognition could be trained on large handwriting datasets, significantly improving accuracy compared to rigid rule-based approaches. In parallel, NLP evolved beyond keyword matching through techniques such as bag-of-words, n-gram modelling, and feature engineering. Systems during this period used measurable features like vocabulary richness, syntactic variety, and answer length; to predict scores. Although these models improved objectivity, they still lacked the capability to interpret semantic meaning or contextual relevance, which are essential for evaluating descriptive responses.

Deep Learning and Modern AI (2010s-Present):

The most transformative advancements arrived with the integration of Deep Learning and neural networks. In the field of OCR, the introduction of Convolutional Neural Networks (CNNs) revolutionized handwriting recognition, enabling systems to learn complex spatial patterns and adapt to diverse writing styles with exceptional accuracy. Concurrently, NLP underwent a major paradigm shift with the emergence of transformer-based architectures such as BERT, RoBERTa, and Sentence Transformers. These models capture contextual relationships between words and sentences, allowing for nuanced understanding of meaning and intent.

2.2 Literature Review

[1] V. R. Anvikar, A. Rajadhyaksha, and R. Deolekar, "Can AI Assistance Aid in the Grading of Handwritten Answer Sheets?" IIT Bombay, 2023.

This work, conducted at IIT Bombay, explores the use of AI not as a replacement for human grading but as an assistant to speed up and improve the evaluation process. The proposed system highlights key terms in students' handwritten answers, helping instructors quickly locate essential information and make more efficient grading decisions. Tested across hundreds of answer scripts, it demonstrated a 31–33% reduction in grading time without compromising grading accuracy. The study positions AI as a collaborative tool that reduces fatigue and bias while ensuring educators maintain control over the final assessment, though it relies heavily on keyword-based methods that sometimes fail to capture the full meaning of student responses.

[2] S. S. Kamble, R. B. Kapoor, M. V. K. Borate, and M. P. L. Mandale, "Design of automated model for inspecting and evaluating handwritten answer scripts – A pedagogical approach with NLP and deep learning," in Springer Conference on Computational Intelligence, pp. 1–10.

This paper presents an advanced model combining deep learning and NLP to evaluate handwritten answer sheets with greater semantic awareness. Instead of depending solely on keyword matching, the system attempts to calculate semantic similarity between student answers and standard answers, thereby mimicking the decision-making process of human evaluators. The model applies NLP techniques to parse and understand meaning, while deep learning methods improve recognition of handwritten text. The study claims promising results in capturing context and meaning, but it highlights challenges related to domain-specific accuracy, computational requirements, and the complexity of making the model interpretable to educators who may need to understand how scores are assigned.

[3] A. R. Joshi and P. K. Deshmukh, "A Survey on Automating Answer Sheet Evaluation Using AI Techniques," International Research Journal of Engineering and Technology (IRJET), vol. 9, no. 7, pp. 890–896, 2022.

This survey paper provides a broad review of existing AI-based approaches for automating answer sheet evaluation, ranging from OCR and NLP models to semantic embedding techniques such as BERT and modern large language models like GPT-4. It highlights the strengths of BERT-based systems in structured and precise evaluation, as well as the creativity and flexibility of large language models that can handle open-ended answers more naturally. The survey underlines the importance of OCR for processing handwritten answer sheets and stresses the growing trend of hybrid frameworks that combine semantic precision with generative understanding. However, the review also notes the challenges of fairness, hallucination risks in LLMs, and the difficulty of integrating such models into standardized academic assessment.

[4] S. Patel and M. Sharma, "Online Subjective Answer Verifying System Using Artificial Intelligence," International Journal of Engineering Research & Technology (IJERT), vol. 10, no. 6, pp. 245–250, 2021.

This paper discusses the design of a subjective answer evaluation system for online examinations, moving beyond traditional multiple-choice grading systems. The proposed system incorporates pre-processing, text recognition, NLP-based similarity measures, and automated scoring to handle descriptive answers submitted in digital form. It emphasizes

scalability and fairness, aiming to reduce evaluator bias while ensuring quicker result processing. The paper highlights the importance of such systems in modern online education, especially with the growing popularity of digital examinations. However, it also recognizes that natural language ambiguity, difficulties in handling synonyms or paraphrased responses, and heavy reliance on pre-processing techniques present obstacles to building a fully reliable system.

[5] M. K. Verma and A. Singh, "Automated Subjective Answer Evaluation Using Machine Learning," International Journal of Engineering Research & Technology (IJERT), vol. 10, no. 5, pp. 555–560, 2021.

This paper proposes a complete pipeline that combines OCR for text extraction, NLP for pre-processing and understanding, and similarity techniques for grading subjective answers. The system reduces the evaluator's workload by up to 90% and even provides automated feedback to students, making it valuable for both teachers and learners. It demonstrates efficiency in descriptive answer evaluation by automating repetitive tasks and speeding up results. Nonetheless, the paper identifies important limitations such as potential inaccuracies in semantic evaluation, dependency on training datasets, and skepticism in real-world adoption due to the need for human oversight and the challenge of aligning AI-driven grading with diverse educational standards.

[6] R. Sharma and N. Gupta, "ANS_CHECKER – Descriptive Answer Checking," International Journal for Scientific Research & Development (IJSART), vol. 8, no. 4, pp. 102–108, 2020.

ANS_CHECKER is presented as a system specifically designed to check descriptive answers using natural language processing techniques. It works by comparing student responses against reference answers, offering a quick and scalable solution for grading descriptive text. The paper highlights its ability to reduce evaluator workload and make assessments more efficient, especially in environments where large numbers of answer sheets must be graded in a short time. However, it is largely dependent on phrasing and struggles with paraphrased or contextually correct answers that do not match the reference wording closely, limiting its effectiveness in scenarios where semantic flexibility is required.

Chapter 3

Limitations in Existing Systems

Despite significant advancements in the field of automated answer sheet evaluation, existing systems continue to face a variety of limitations that hinder their widespread adoption in real-world academic settings. Most solutions developed so far are highly dependent on the accuracy of OCR tools, which often struggle with poor image quality, diverse handwriting styles, and noisy backgrounds. Even when text is extracted correctly, many models rely heavily on keyword matching or surface-level similarity measures, which fail to capture the true semantic meaning of student responses. This often results in unfair grading, especially when answers are expressed in alternative but contextually correct phrasing.

Furthermore, while some systems attempt to incorporate NLP and deep learning, they are frequently constrained by domain-specific datasets, high computational requirements, or limited interpretability, making them difficult for educators to trust and adopt. Online evaluation systems have also grown in importance, but their effectiveness is often reduced by the ambiguity of natural language, the lack of robust paraphrase handling, and challenges in integrating with large-scale digital platforms. These limitations emphasize the need for a more sophisticated approach that combines accurate text recognition, semantic analysis, and scalable architecture to provide reliable, unbiased, and efficient evaluation.

Table 3.1: Summary of limitations of existing systems

Title	Conference / Journal Details	Key Points	Improvements Proposed	Citation
Can AI Assistance Aid in the Grading of Handwritten Answer Sheets?	IIT Bombay, 2023 (workshop paper)	AI-assisted keyword highlighting reduced grading time by 31–33%; kept human in the loop	Move beyond keyword reliance, integrate semantic evaluation	[1]
Design of Automated Model for Inspecting and Evaluating Handwritten Answer Scripts – A Pedagogical Approach with NLP and Deep Learning	Springer Conference, 2022	Combines NLP + Deep Learning to assess semantic similarity; mimics human-like grading	Enhance interpretability, improve adaptability across domains	[2]
A Survey on Automating Answer Sheet Evaluation Using AI Techniques	IRJET, 2022	Surveys OCR, NLP, BERT, GPT-4 techniques; compares semantic precision with creative grading	Develop hybrid frameworks balancing structure and creativity	[3]
Online Subjective Answer Verifying System Using Artificial Intelligence	IJERT, 2021	AI-based descriptive answer grading for online exams; reduces bias, scalable for large cohorts	Better handling of paraphrasing, synonyms, and ambiguous responses	[4]
Automated Subjective Answer Evaluation Using Machine Learning	IJERT, 2021	OCR + NLP + ML pipeline; reduces workload up to 90%; automated feedback to students	Improve semantic accuracy, dataset independence, adoption in education	[5]
ANS_CHECKER – Descriptive Answer Checking	IJSART, 2020	NLP-based answer checking; scalable and reduces manual workload	Reduce dependence on phrasing; improve paraphrase recognition	[6]

Chapter 4

Problem Statement & Objectives

4.1 Problem Statement

The process of evaluating answer sheets in academic institutions has always been an integral part of the education system, but it continues to present significant challenges. Traditionally, evaluation is carried out manually by teachers or examiners who are required to read through each answer carefully, interpret the student's understanding, and allocate marks accordingly. While this system allows for flexibility in judgment, it is also highly time-consuming, especially in scenarios involving large class sizes or mass examinations. Manual evaluation often results in inconsistencies, as different evaluators may assign different marks for the same response due to varying levels of strictness, fatigue, or unintentional human bias.

Moreover, with the increasing emphasis on online and remote education, the limitations of manual grading are becoming more apparent. Automated methods exist for multiple-choice questions through Optical Mark Recognition (OMR), but descriptive and subjective answers remain difficult to assess. Variations in handwriting, differences in phrasing, and the need for deeper semantic understanding further complicate the problem. Existing automated approaches are either too rigid, relying on exact keyword matching, or too experimental, struggling to balance fairness and creativity in student responses

4.2 Objectives

In order to address the challenges identified in the problem statement, this project sets out the following objectives:

1. **To automate the process of answer sheet evaluation:** Develop a system capable of processing scanned or photographed answer sheets and eliminating the need for manual checking.
2. **To extract text accurately using OCR:** Implement image pre-processing techniques such as binarization, denoising, and skew correction to improve text recognition quality, thereby ensuring that even noisy or unclear scripts can be interpreted with reasonable accuracy.
3. **To incorporate natural language understanding:** Apply NLP and machine learning techniques to evaluate descriptive answers by comparing them with model answers, ensuring that marks are awarded based on meaning rather than rigid keyword presence.
4. **To ensure consistency and fairness in evaluation:** Design the system to minimize the effects of human bias, fatigue, and variability across evaluators, thus providing a standardized approach to grading.
5. **To reduce time and effort for teachers and institutions:** Enable large-scale automated evaluation to save valuable teaching time, allowing educators to focus more on mentoring and guiding students rather than on repetitive administrative work.
6. **To build a foundation for future enhancements:** Keep the system flexible enough to accommodate advanced features such as diagram recognition, handwriting adaptation through deep learning models, and integration into existing digital examination platforms.
7. **To provide timely and constructive feedback:** Develop the system to not only assign marks but also generate meaningful feedback that can help students understand their mistakes and improve their writing and conceptual clarity.
8. **To ensure scalability and adaptability across subjects:** Design the system so it can be applied to a wide range of academic disciplines and question types, ensuring flexibility and robustness for different educational contexts.

Chapter 5

Proposed System

The framework operates as a multi-stage pipeline, where each stage contributes to transforming raw handwritten responses into meaningful evaluation results. Initially, scanned or photographed answer sheets undergo image pre-processing, including grayscale conversion, noise removal, skew correction, and binarization, to enhance clarity and optimize them for text recognition. This stage ensures that even low-quality or noisy inputs can be effectively processed. The cleaned images are then passed through Optical Character Recognition (OCR) or Handwritten Text Recognition (HTR) tools such as Tesseract or EasyOCR, which convert the handwritten content into machine-readable text. Once digitized, the extracted answers are further refined through text pre-processing and then analysed in the NLP-based evaluation module, where semantic similarity with model answers is computed to capture the actual meaning conveyed by the student. A dedicated scoring logic module integrates semantic similarity measures, keyword presence, and optional diagram evaluation to generate consistent, unbiased, and transparent marks.

The system goes beyond simple automation by ensuring fairness and efficiency in the evaluation process. Its modular design provides a foundation for future enhancements, including diagram recognition through deep learning, adaptation to diverse handwriting styles, and seamless integration with digital examination platforms. Collectively, these features highlight the system’s potential not only as an automated grading tool but also as a step toward creating intelligent educational technologies that support both teachers and students

5.1 System Architecture

The architecture of the proposed system is designed as a modular pipeline, enabling the transformation of raw handwritten answers into objective evaluation results. As illustrated in Figure X, the process begins with the input stage, where scanned or photographed student and model answer sheets are provided to the system. These inputs are processed through a text extraction module that employs Optical Character Recognition (OCR) or Handwritten Text Recognition (HTR) tools to digitize the handwritten responses. The extracted text is then normalized through pre-processing operations, including tokenization, stop word removal, stemming, and lemmatization, which ensure that irrelevant components are eliminated and only meaningful linguistic features are retained. Subsequently, the text is passed to the vectorization module, where advanced embedding models such as TF-IDF, Word2Vec, or BERT are used to represent words and sentences in numerical form.

The evaluation stage follows, where three critical dimensions are measured: semantic similarity, keyword overlap, and conceptual relevance. This ensures that answers are not judged solely on exact phrasing but also on their ability to convey the correct meaning. Finally, the evaluation process employs a weighted scoring logic, assigning 70% importance to semantic similarity, 10% to keyword overlap, and 20% to word count.

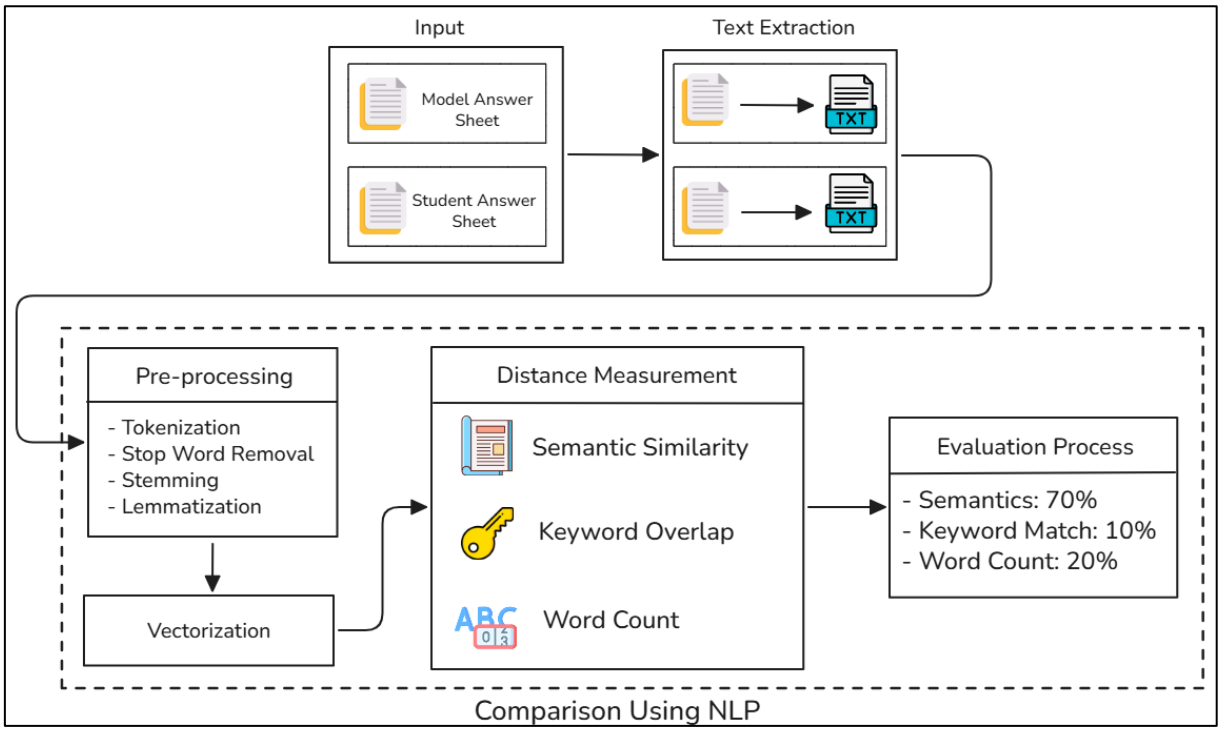


Figure 5.1: Overall system architecture & workflow

Description of figure 5.1:

- **Input Stage:** Scanned/photographed student and model answer sheets are acquired.
- **Text Extraction (OCR/HTR):** Handwritten text is digitized using Google Gemini Vision API.
- **Text Pre-processing:** Operations include tokenization, stop word removal, stemming, and lemmatization. Ensures text is normalized and ready for analysis.
- **Vectorization:** Converts text into numerical embeddings using TF-IDF, Word2Vec, or BERT.
- **Distance Measurement:** Compares student and model answers on three parameters:
  - Semantic similarity
  - Keyword overlap
  - Conceptual relevance
- **Evaluation Process:** Weighted scoring logic applied:
  - 70% semantics
  - 10% keyword match
  - 20% word count
- **Output:** Generates marks and lays the foundation for unbiased, scalable evaluation.

5.2 System Modules

To provide a clearer understanding of the workflow, the system can be divided into the following functional modules:

- **Image Pre-processing:** Enhances the quality of scanned answer sheets by applying operations such as grayscale conversion, noise removal, skew correction, and binarization. This stage optimizes the document for accurate OCR recognition.
- **OCR/HTR Module:** Converts handwritten or printed responses into digital text. Tools like Tesseract or EasyOCR are employed to handle the diversity in handwriting styles and formats.
- **Text Pre-processing:** Normalizes the extracted text by applying NLP techniques such as tokenization, stop word removal, stemming, and lemmatization. These processes ensure that only meaningful content is preserved for evaluation.
- **Vectorization:** Converts text into numerical vectors using models like TF-IDF, Word2Vec, or BERT embeddings. This enables semantic comparison between student and model answers.

- **Distance Measurement:** Calculates semantic similarity, keyword overlap, and conceptual relevance. This module forms the core of the evaluation by bridging the gap between surface-level word matching and deeper semantic understanding.
- **Evaluation Process:** Applies a weighted scoring logic (70% semantics, 10% keyword matching, 20% word count) to generate objective marks. This approach ensures fairness by balancing meaning, terminology, and contextual alignment.
- **Result and Feedback:** Produces the final marks for each answer. The system can also be extended to generate constructive feedback to guide students in improving their responses.

5.3 Workflow Explanation

The workflow of the proposed system begins with the acquisition of scanned or photographed answer sheets, which serve as the primary input. These images first undergo pre-processing operations such as grayscale conversion, noise removal, skew correction, and binarization to enhance their quality and optimize them for recognition. The refined images are then passed through the OCR/HTR module, where handwritten or printed responses are extracted and converted into digital text. Once the text is obtained, it is subjected to a series of NLP-based pre-processing steps, including tokenization, stop word removal, stemming, and lemmatization, to standardize the content and eliminate redundancy. The cleaned text is subsequently transformed into numerical embeddings through vectorization techniques such as TF-IDF, Word2Vec, or BERT, enabling the system to capture semantic meaning beyond surface-level words. Following this, the evaluation module compares the student's response with the model answer across three key dimensions: semantic similarity, keyword overlap, and conceptual relevance. The results are then processed by the scoring logic, which applies a weighted scheme:70% semantics, 10% keyword overlap, and 20% word count; to generate a fair and consistent score. Finally, the system produces the output in the form of marks, with the potential to extend toward providing constructive feedback, thereby ensuring that evaluation is both efficient and unbiased.

5.4 Advantages of Proposed System

- **Time Efficiency:** Significantly reduces the manual workload for teachers in large-scale assessments.
- **Semantic Understanding:** Rewards students for conveying correct meaning, even if expressed differently.
- **Future Readiness:** Provides a foundation for enhancements such as diagram recognition, handwriting adaptation, and integration into e-learning platforms.

Chapter 6

Experimental Setup

This chapter presents the complete configuration and setup used for implementing and testing the system. The experimental environment is divided into two major parts: Frontend Setup and Backend Setup; both of which play distinct yet complementary roles in the project. The backend is responsible for text extraction, pre-processing, and intelligent evaluation, while the frontend provides a user-friendly interface for uploading answer sheets and viewing results. The integration of these components ensures an efficient, scalable, and automated approach to evaluating handwritten or typed answers.

6.1 Frontend Setup

The frontend of Answer Check serves as the user interface that allows teachers and students to interact with the system. It was developed using React.js to ensure a fast, modular, and responsive web experience.

Technologies Used:

- **React.js:** for component-based UI development.
- **HTML5 & CSS3:** for structuring and styling the web pages.
- **JavaScript (ES6):** for dynamic content and logic handling.
- **Axios:** for handling API requests to the backend.
- **Vite:** for faster build and development server setup.

UI Features:

- Simple and user-friendly design.
- Drag-and-drop or click-to-upload functionality.
- Displays extracted text and final score.
- Responsive layout supporting desktops and tablets.

6.2 Backend Setup

The backend forms the core logic of the Answer Check system, handling file uploads, OCR extraction, NLP evaluation, and score computation. It is implemented using FastAPI for fast and efficient API development in Python.

Technologies Used:

- **FastAPI:** for creating RESTful APIs.
- **Python 3.10+:** core programming language.
- **Sentence Transformers:** for semantic similarity.
- **TextBlob:** for tone/sentiment analysis.
- **NLTK:** for text pre-processing (tokenization, stopword removal, lemmatization).
- **pdf2image:** for converting PDF pages to images (for OCR).
- **Pillow:** for image processing.
- **Gemini Vision API:** for OCR text extraction from handwritten inputs.

Backend Functional:

- Handles multiple file formats (PDF, DOCX, PNG, JPG).
- Performs fall back to OCR if text extraction fails.
- Computes three-level evaluation:
  - Keyword similarity
  - Semantic similarity
  - Tone alignment
- Returns JSON output with detailed scoring.

6.3 Development Environment:

The development environment was configured to ensure smooth integration and testing of all modules.

Environment Details:

- **IDE:** Visual Studio Code
- **API Testing:** Postman / REST Client
- **Version Control:** Git (optional)
- **Operating System:** Windows 10 / 11 (64-bit)

The experimental setup of the Answer Check system provided the foundation for implementing and testing the proposed model efficiently. The frontend, developed using React.js, ensures an intuitive and responsive interface for users to upload and evaluate answers. The backend, powered by FastAPI and various NLP and OCR libraries, performs the core processing, text extraction, and scoring with accuracy and scalability.

A well-configured development environment with the required hardware and software tools enabled smooth integration between the modules, ensuring reliable system performance during evaluation. This setup successfully supports the end-to-end workflow; from input upload to result generation; validating the effectiveness of the proposed architecture in automating subjective answer evaluation.

Chapter 7

Results & Discussion

This chapter presents the outcomes of the Subjective Handwritten Answer Evaluation system and analyses its performance in various stages, including text extraction, pre-processing, semantic similarity evaluation, and final scoring. The obtained results demonstrate the efficiency, consistency, and scalability of the developed model compared to manual evaluation methods.

7.1 Results

The project outcomes confirm the model's performance in text extraction, pre-processing, semantic similarity evaluation, and final scoring. The results obtained demonstrate the efficiency, consistency, and scalability of Answer Check compared to traditional manual methods. The integration of the Gemini Vision API proved crucial for accurate recognition of diverse handwriting styles. The use of TF-IDF vectorization and cosine similarity for semantic comparison successfully enabled the system to reward students based on meaning and context, rather than being strictly dependent on phrasing or exact keyword matches.

The weighted grading model (70% semantics) ensures that the system provides a comprehensive assessment. The automated evaluation aligns closely with human grading for most responses, validating the feasibility of using NLP for subjective assessments. By minimizing human bias and ensuring consistency across large batches, the system lays the groundwork for objectivity and transparency in educational evaluation.

7.2 Demonstration

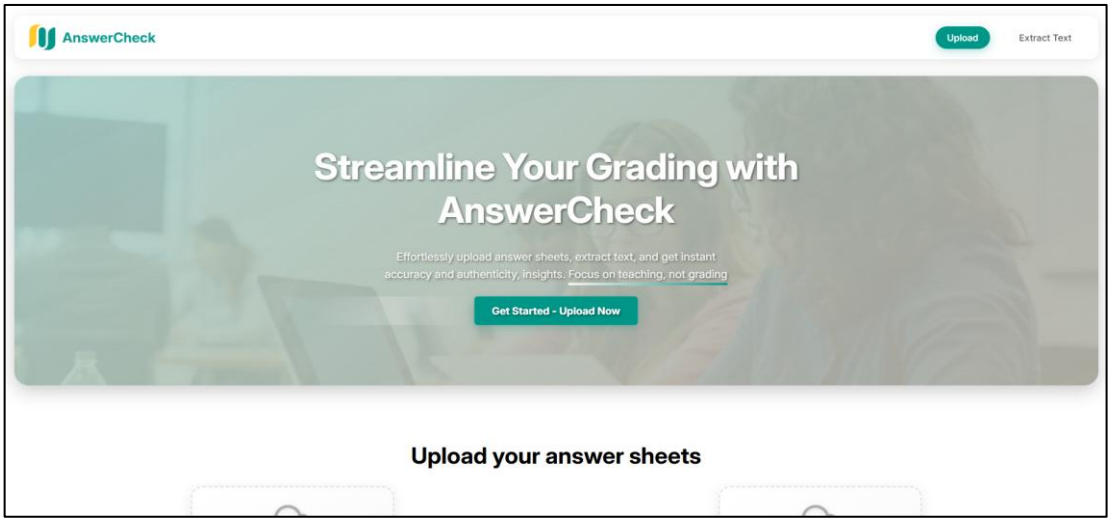


Figure 7.1: Landing page of Answer Check

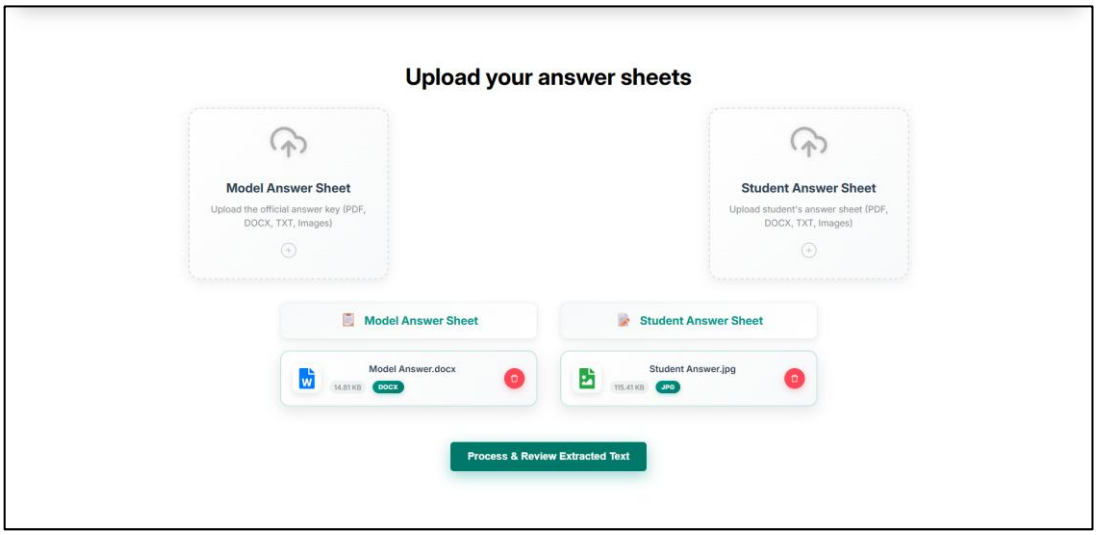


Figure 7.2: Upload page after selecting model and student answer files

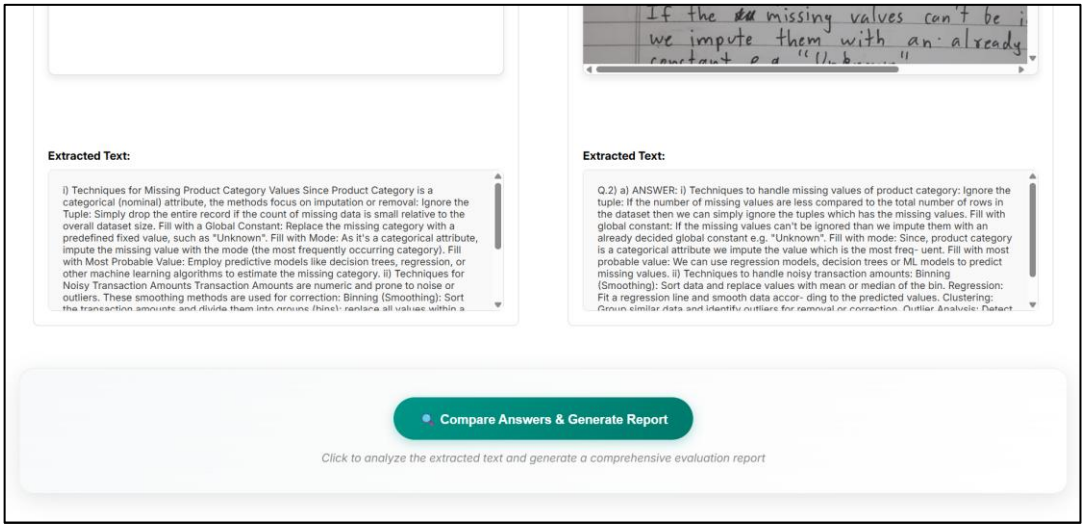


Figure 7.3: Extracted text from model and student answers

i) Techniques for Missing Product Category Values

Since **Product Category** is a categorical (nominal) attribute, the methods focus on imputation or removal:

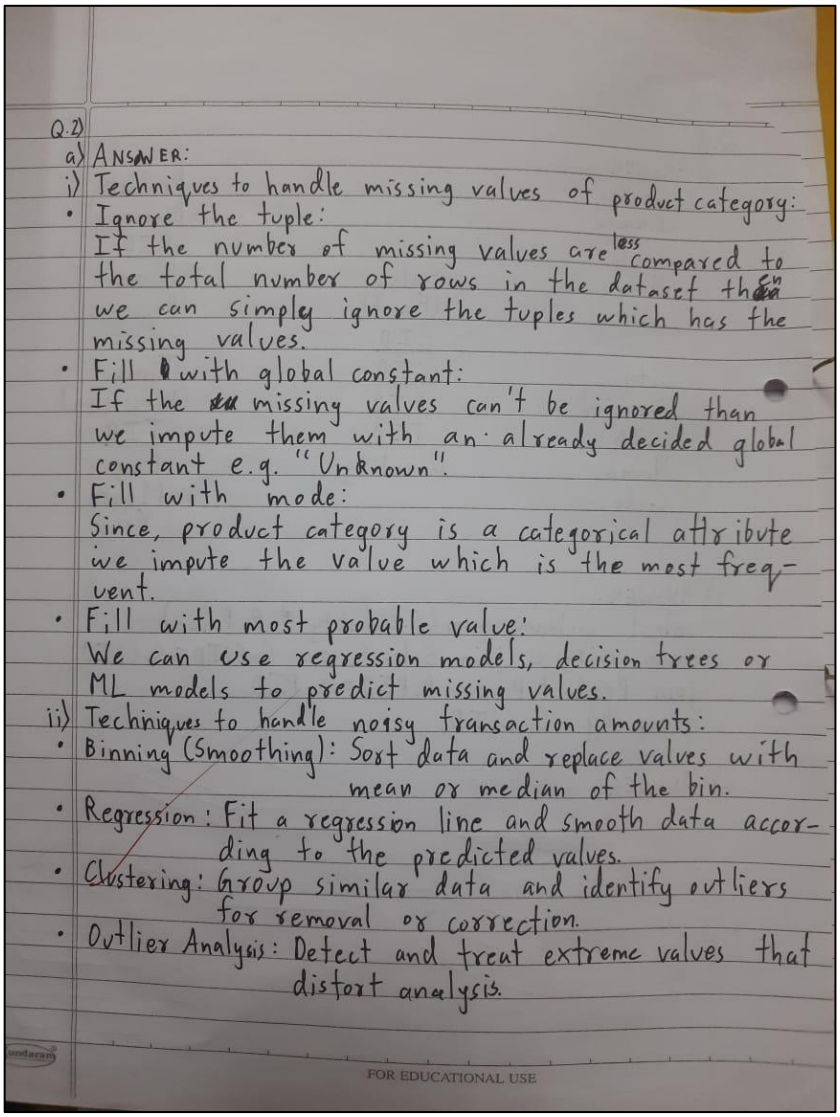
- **Ignore the Tuple:** Simply drop the entire record if the count of missing data is small relative to the overall dataset size.
- **Fill with a Global Constant:** Replace the missing category with a predefined fixed value, such as "Unknown".
- **Fill with Mode:** As it's a categorical attribute, impute the missing value with the **mode** (the most frequently occurring category).
- **Fill with Most Probable Value:** Employ **predictive models** like decision trees, regression, or other machine learning algorithms to estimate the missing category.

ii) Techniques for Noisy Transaction Amounts

Transaction Amounts are numeric and prone to noise or outliers. These smoothing methods are used for correction:

- **Binning (Smoothing):** Sort the transaction amounts and divide them into groups (bins); replace all values within a bin with either the **bin mean** or **bin median**.
- **Regression:** Use **regression analysis** to fit a trend line to the data, and replace the noisy amounts with the corresponding predicted values from the line.
- **Clustering:** Group similar transaction amounts into clusters; data points that are far from the cluster centers are often identified as noise or **outliers** for subsequent treatment or removal.
- **Outlier Analysis:** Specifically detect and handle **extreme values** (outliers) that could skew analysis results. This often involves statistical checks, such as the 1.5×IQR rule.

Figure 7.4: Content of "Model Answer.docx"



Q.2)

a) ANSWER:

i) Techniques to handle missing values of product category:

- Ignore the tuple:
  If the number of missing values are less compared to the total number of rows in the dataset then we can simply ignore the tuples which has the missing values.
- Fill with global constant:
  If the missing values can't be ignored than we impute them with an already decided global constant e.g. "Unknown".
- Fill with mode:
  Since, product category is a categorical atribute we impute the value which is the most frequent.
- Fill with most probable value:
  We can use regression models, decision trees or ML models to predict missing values.

ii) Techniques to handle noisy transaction amounts:

- Binning (Smoothing): Sort data and replace values with mean or median of the bin.
- Regression: Fit a regression line and smooth data according to the predicted values.
- Clustering: Group similar data and identify outliers for removal or correction.
- Outlier Analysis: Detect and treat extreme values that distort analysis.

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Figure 7.5: Content of "Student Answer.jpg"

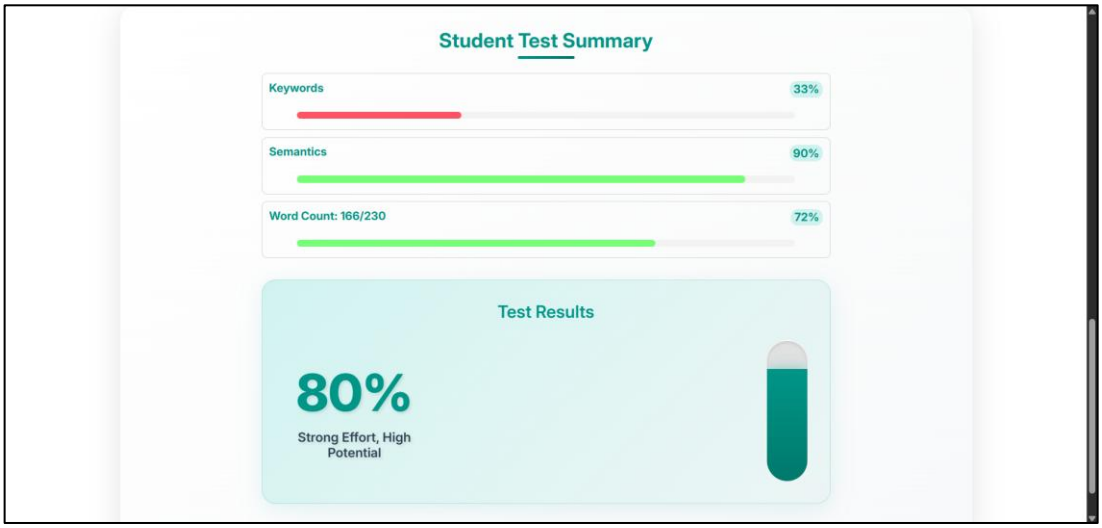

Figure 7.6: Evaluation of student answer compared to the model answer

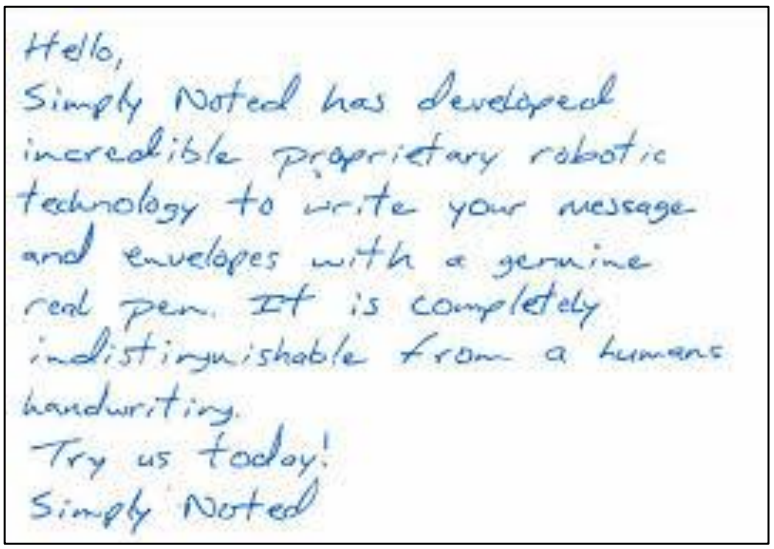

Figure 7.7: Random image with text

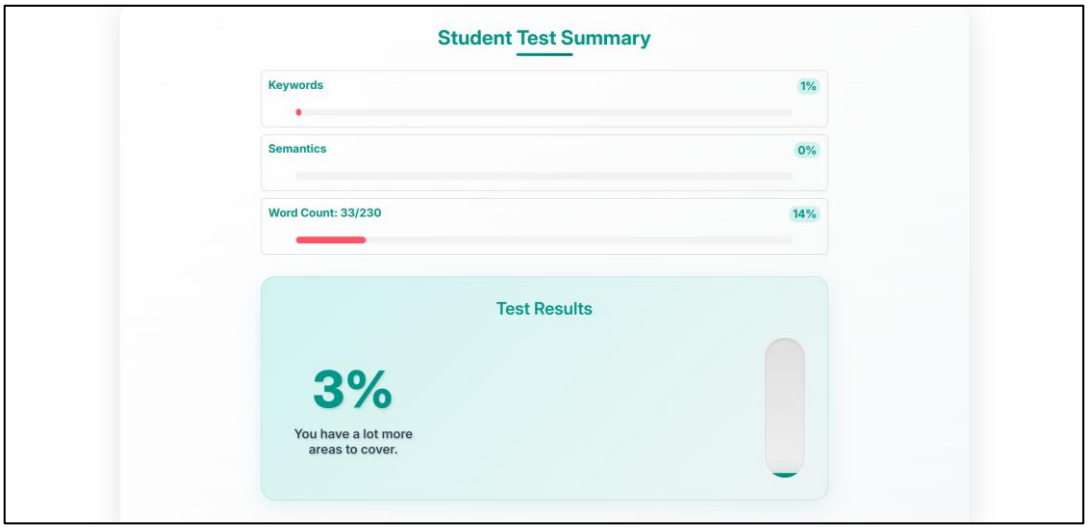

Figure 7.8: Evaluation of the random image compared to the model answer

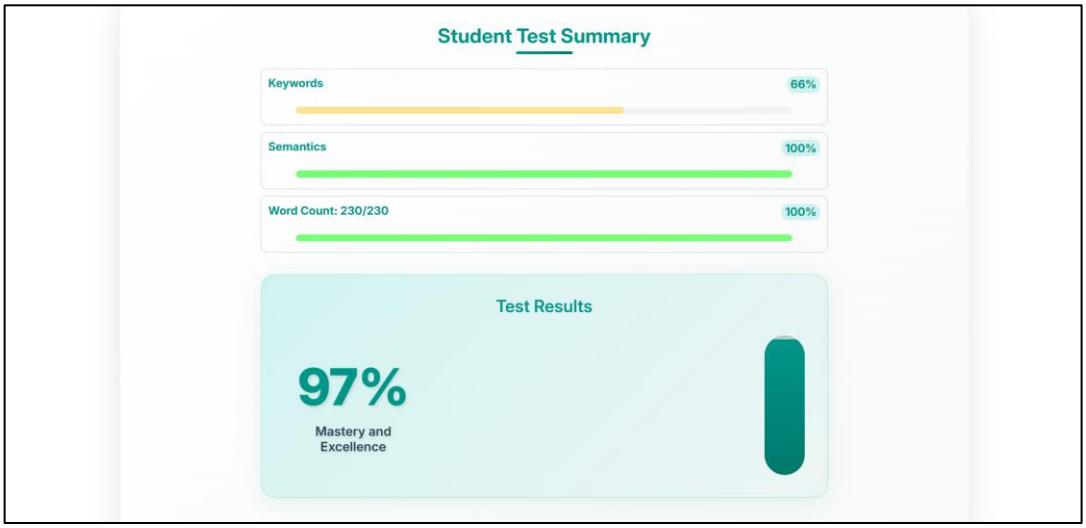

Figure 7.9: Evaluation of the model answer compared to the model answer

7.3 Discussion

The results of the "Subjective Handwritten Answer Evaluation using NLP" project, under the system name Answer Check, validate the effectiveness of integrating Optical Character Recognition (OCR) and Natural Language Processing (NLP) for automated academic evaluation. The discussion focuses on the system's key achievements and how its design choices address the limitations of traditional and existing automated grading methods.

Key Performance Outcomes

- **Accurate Text Extraction:** The core challenge of evaluating handwritten answers; digitizing the text; was successfully addressed by incorporating the Gemini Vision API for text extraction. This integration proved crucial for achieving accurate recognition of diverse handwriting styles, a significant improvement over earlier, more rigid rule-based or less adaptive OCR systems.
- **Semantic-Centric Grading:** Unlike many conventional systems that heavily rely on simple keyword matching, Answer Check employed TF-IDF vectorization and cosine similarity to compare the student's response with the model answer semantically. This emphasis on contextual understanding and meaning ensures that the system rewards students even when their answers are phrased differently but are still contextually correct.
- **Weighted Scoring for Fairness:** The final score is derived from a weighted grading model, which allocates 70% importance to semantic similarity, 10% to keyword match, and 20% to word count (as shown in Figure 7.6, 7.8 & 7.9). This multi-metric approach ensures a comprehensive and balanced assessment that closely mirrors human evaluation standards by balancing meaning, terminology, and contextual alignment.

- **Consistency and Objectivity:** Experimental results confirm that the automated scores align closely with human grading for most responses. This validates the system's feasibility and its ability to minimize the effects of human bias, fatigue, and variability across evaluators, leading to a standardized and transparent grading process.

Overcoming Existing Limitations:

The proposed system directly addresses several limitations identified in the literature and existing systems (as summarized in Table 3.1):

- **Paraphrasing and Ambiguity:** By prioritizing semantic evaluation (70% weight), the system moves beyond simple keyword reliance and effectively handles paraphrased or synonymous responses, an issue that limited earlier NLP-based tools like ANS_CHECKER.
- **Lack of Interpretability:** While advanced deep learning models can be complex to interpret, Answer Check provides a breakdown of scores based on Keywords, Semantics, and Word Count (Figure 6), which enhances transparency and allows educators to understand how marks are assigned.
- **Efficiency and Scalability:** The backend, built with Python and FastAPI, ensures efficient handling of processing and API logic, which, combined with the automated workflow, significantly reduces the manual workload and grading time for teachers. This scalability makes the system suitable for large-scale academic assessments.

In essence, Answer Check successfully demonstrates the potential of combining cutting-edge AI and NLP to modernize academic evaluation, offering a reliable, objective, and efficient alternative to manual grading.

Chapter 8

Conclusion & Future Scope

8.1 Conclusion

The project “Subjective Handwritten Answer Evaluation using NLP”, developed under the system name Answer Check, represents a practical implementation of modern Artificial Intelligence techniques in the education domain. The system effectively automates the evaluation of descriptive and subjective answers by integrating Optical Character Recognition (OCR) for text extraction and Natural Language Processing (NLP) for semantic understanding and grading.

In traditional examination systems, evaluating subjective answers is a time-consuming and often inconsistent process due to human bias, fatigue, and workload. The proposed system addresses these limitations by introducing a consistent, data-driven approach to answer evaluation. By combining multiple evaluation criteria: keyword similarity, semantic meaning, and tone analysis; the system provides a more comprehensive assessment of student responses. Unlike simple keyword-based systems, Answer Check interprets the context and intent behind a student’s answer, allowing fairer and more accurate grading.

The backend, implemented using FastAPI, efficiently handles file processing, OCR integration, and NLP computations, while the React.js-based frontend ensures smooth user interaction and visualization of results. The modular design allows scalability and easy maintenance, ensuring the system can evolve with future advancements in AI and NLP technologies.

Experimental results indicate that the automated evaluation aligns closely with human grading for most responses, demonstrating the feasibility of using NLP for subjective assessments. Teachers can now focus more on instructional activities rather than repetitive grading tasks, while students can receive faster feedback to improve their learning outcomes.

Beyond automation, Answer Check contributes to promoting objectivity, transparency, and scalability in educational evaluation. By minimizing human bias and ensuring consistency across large batches of answer sheets, the system lays the groundwork for a new era of digital education where fairness and efficiency coexist.

8.2 Future Scope

Although the system performs efficiently in its current form, there are several areas for enhancement and expansion to make it more comprehensive and intelligent.

1. **Diagram Recognition:** Extend the system to evaluate diagrams and figures using image classification and object detection techniques.
2. **Advanced Semantic Understanding:** Incorporate transformer-based deep learning models like BERT or RoBERTa for more nuanced understanding of complex and descriptive answers.
3. **Multi-Language Support:** Add support for regional languages by integrating multilingual OCR and NLP models.
4. **Adaptive Handwriting Models:** Train custom handwriting recognition models that adapt to individual writing styles for improved OCR accuracy.
5. **Integration with Online Examination Portals:** Combine the system with digital exam management platforms to automatically evaluate uploaded answer sheets in real time.
6. **Feedback Generation:** Include automated feedback modules that highlight key strengths and areas for improvement for each student.
7. **Cloud-Based Scalability:** Deploy the system on a cloud environment (e.g., Google Cloud or AWS) for large-scale processing of multiple answer sheets simultaneously.

In conclusion, this project lays the groundwork for the future of AI-driven educational assessment systems. By automating handwritten answer evaluation, it not only addresses existing challenges such as subjectivity and inefficiency but also paves the way for a more data-driven, transparent, and scalable approach to grading. With continued development and optimization, systems like this have the potential to revolutionize the way academic evaluations are conducted, benefiting both educators and students alike.

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